IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Jonathan Sargent
04-18-2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data collection with API and webscraping from Wikipedia
  - Data cleaning and formatting
    - Data analysis using SQL
    - Data wrangling
    - Standardized the data
  - Visualizations of data with folium and plotly dash
    - Mapped launch sites and success rates
    - Built an interactive dashboard with plotly dash
    - Data analysis for Visualization
  - Implemented SVM, Linear Regression, Decision Trees, and K-Nearest Neighbors
    - Evaluated models using test data accuracy
  - Python machine learning for predictive analysis
    - Data determined landing success rate accuracy

- Summary of all results
  - Used data analysis, Interactive analytics, and predictive analysis to identify influencing factors
    - For landing success
  - SVM and K-Nearest neighbors: 83.3% accuracy.
  - Decision Tree: 88.8% accuracy
  - Launch site and payload mass impact landing success
  - SpaceX Falcon 9 is predicted to land safely
  - Decision Tree model is the most effective predictor

Introduction

- Project background and context:
  - Using collected data to predict if the Falcon9 can land successfully. The information gathered can be used to increase the confidence of successful future launches in a safe and cost effective manner. SpaceX advertises a low cost compared to other providers, because they reuse the first stage of the rocket. Launch costs can be estimated by accurately predicting landing success

- Problems you want to find answers
  - Finding out if SpaceX can land successfully within the constraints of budget, mass, orbit type, etc.; all while meeting their budget constraints of launching rockets through time.
    - Determine specific for a first stage successful landing of the Falcon 9
    - Accurately predict the landing outcome using machine learning models
    - Determine which machine learning model performs the best in predicting the landing success

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Using the SpaceX API
    - Provided detailed records of Falcon 9 launches
      - Launch dates
      - Payloads
      - outcomes
- Perform data wrangling:
  - Processed with data filtering
  - Handling missing values
  - Standardizing formats
  - Ensuring consistency
  - Key features were extracted
- Perform exploratory data analysis (EDA) using visualization and SQL:
  - Used matplotlib and Seaborn for launch success rates
  - Executed SQL queries to derive insights and answer specific questions

Methodology Continued

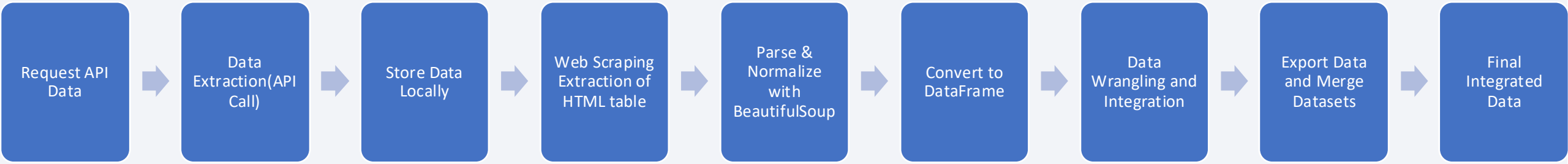
- Perform interactive visual analytics using Folium and Plotly Dash:
  - Created interactive maps displaying launch sites and outcomes
  - Created a plotly dash application with a dropdown and sliders to analyze launch success rates and payload mass ranges
- Perform predictive analysis using classification models:
  - How to build, tune, and evaluate classification models using logistic regression, support vector machines, decision trees, and k-nearest neighbors
  - Evaluated models for accuracy. Determined best performing model for predicting landing success

My Github URL:

https://github.com/T-afk-design/courseraCapstone

Data Collection

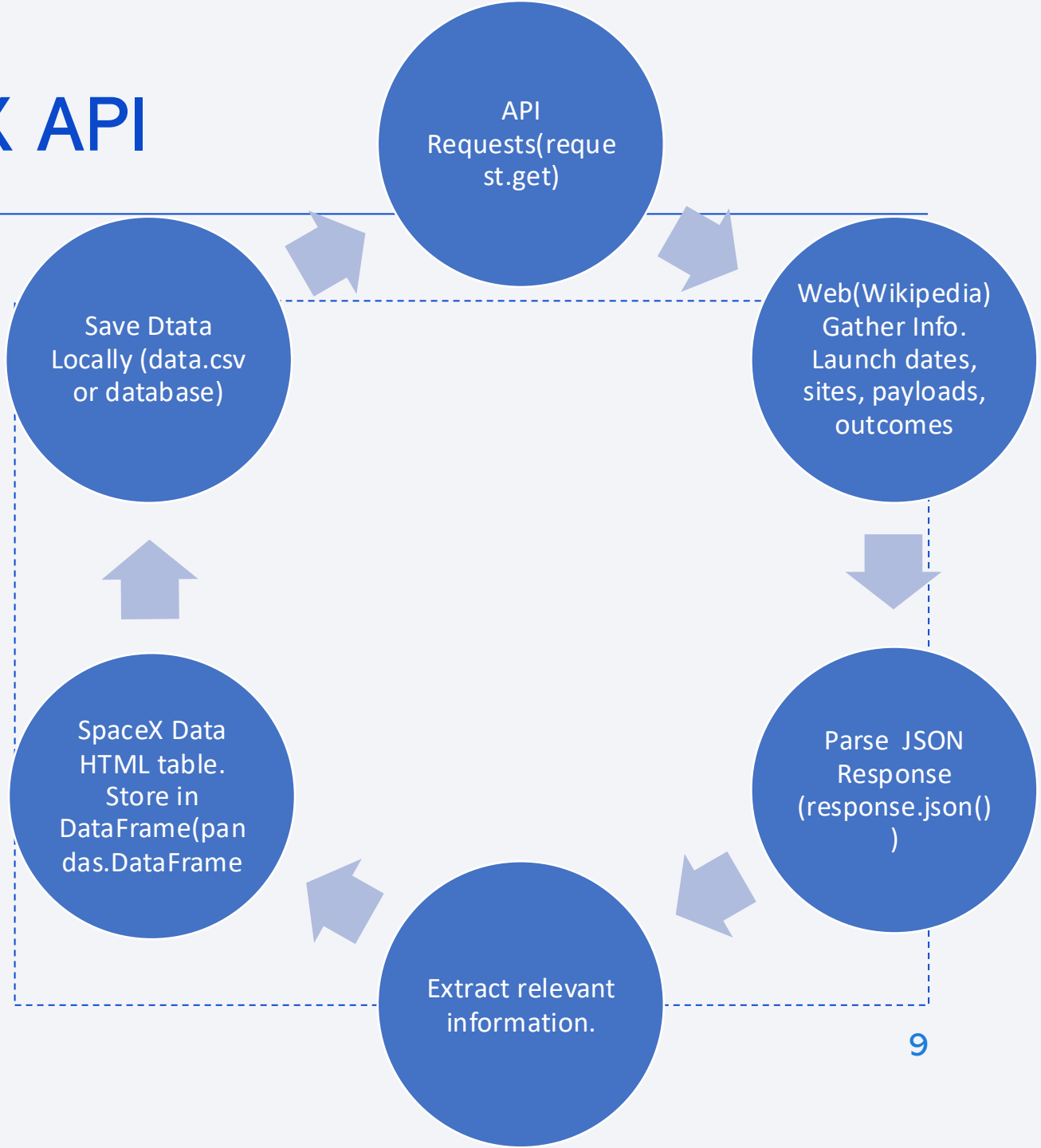
- Data was collected with SpaceX API Request, Web Scraping Wikipedia, Data Integration
  - Numpy and pandas libraries used to format data
- Data collection process used:



Data Collection – SpaceX API

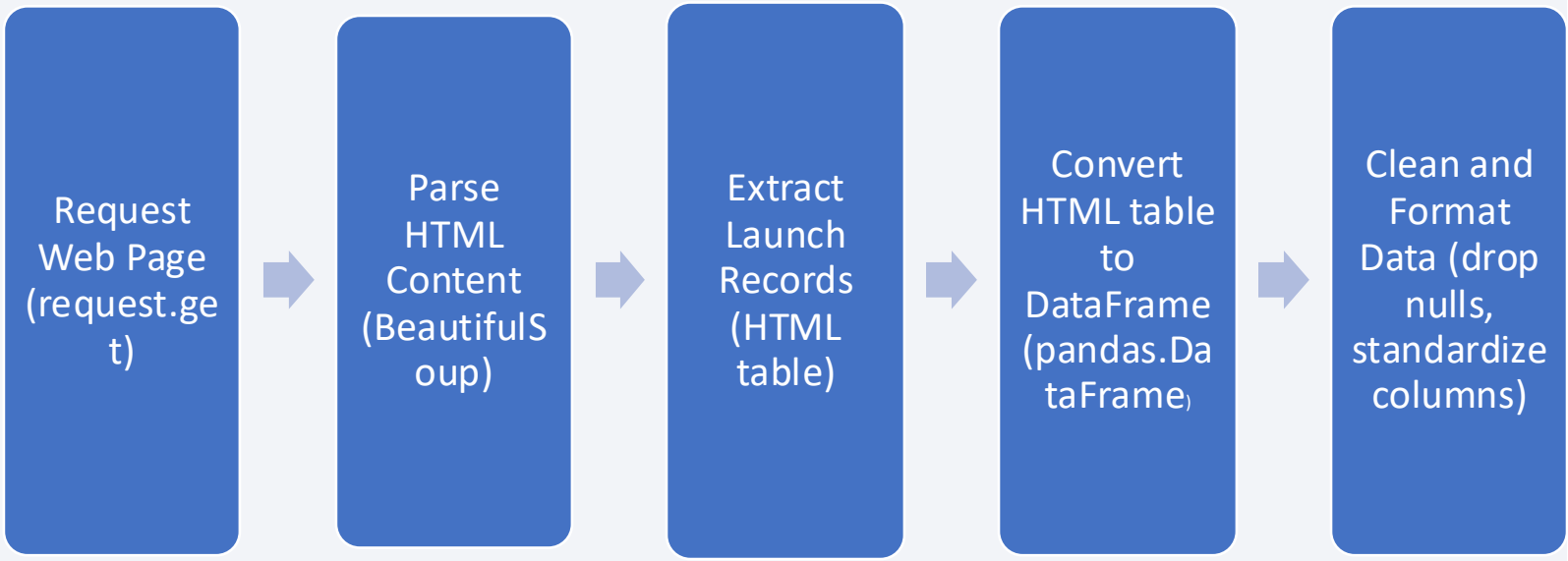
- Step 1: Initiate API Request
  - Use Pythons ‘requests’ library to connect to the SpaceX API.
- Step 2: Parse API Response
  - Convert API response from JSON to a Python dictionary
  - Extract relevant fields, launch date, site, payload mass, booster type, and outcome
- Step 3: Store Data Locally
  - Save extracted data into a pandas DataFrame
  - Store the DataFrame locally for further processing
  - Github URL:

https://github.com/T-afk-design/courseraCapstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- First, initiate web scraping by using python's requests library to get HTML data from the target URL
- Second, Parse the HTML content using BeautifulSoup and extract the HTML table with the Falcon 9 launch records
- Third, convert the extracted data into a pandas DataFrame, then clean and format the data for consistency.
- Github URL: https://github.com/T-afk-design/courseraCapstone/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

- Data wrangling involves cleaning, transforming, and organizing raw data into a structured format suitable for analysis.
  - Data cleaning fills or removes missing values in the dataset
  - Rows/columns with excessive missing data are dropped
- Data Transformation makes datasets readable and consistent
  - Data is formatted
  - Text is standardized
  - Normalize/scale numerical features for clarity and consistency
  - Create new features from existing data
- Data is Integrated
  - Datasets are merged from different sources via API and webscraping
  - Consistent column names and formats are implemented once datasets are merged
- Data is Validated
  - Accuracy and consistency of datasets is verfified
  - Duplicate records are removed

Github URL:

https://github.com/T-afk-design/courseraCapstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **Overview:**
- Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to
- understand the data's distribution, identify patterns, and uncover relationships between variables.
- **Charts Plotted:**
- **1. Histograms:**
- - **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight
- number.
- - **Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.
- **2. Bar Charts:**
- - **Purpose**: Used to compare categorical variables such as launch outcomes (success/failure) across different categories like
- launch sites or rocket types.
- - Why: Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.
- **3. Line Charts:**
- - **Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- - **Why**: Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

EDA with Data Visualization

- 4**. Scatter Plots:**
- - **Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.
- - Why: Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.
- 5**. Heatmaps:**
- - Purpose: Used to visualize correlation matrices between multiple numerical variables.
- - Why: Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding
- multicollinearity.
- 6**. Box Plots:**
- - Purpose: Used to display the distribution of numerical data through their quartiles.
- - Why: Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

- Github URL: https://github.com/T-afk-design/courseraCapstone/blob/main/edadataviz.ipynb

EDA with SQL

- Summarize the SQL queries you performed
  - Display the name of each launch site
  - Display average payload mass carried by booster F9 v1.1
  - Display total number of successful and failure mission outcomes
  - Display total payload mass carried by boosters launched by NASA
  - Nested queries to calculate derived metrics
    - Average payload mass per launch site
  - Sorted data to identify trends and outliers
  - Filtered data to focus on specific launch outcomes
  - Joined tables and combined datasets for comprehensive analysis
  - Calculated success rate by launch site and rocket type
  - And more

- GitHub URL: https://github.com/T-afk-design/courseraCapstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map Objects Created

Markers:

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location
- where SpaceX launches have occurred.

Circles:

- Added circles around launch sites to visually represent
- proximity zones.
- Circles help visualize the areas around launch sites that
- might influence operational decisions.

Lines:

- Drew lines to connect launch sites with their proximities

or other relevant locations.

  - A polyline showing the distance to the coastline

- Lines provide spatial context and connections between
- different points of interest related to launches.
- GitHub URL: https://github.com/T-afk-design/courseraCapstone/blob/main/lab_jupyter_launch_site_location.ipynb

Reasons for Adding Objects

Markers:

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted
- launches geographically.

Circles:

- Illustrates the potential impact zones around launch
- sites.
- Provides a visual representation of safety perimeters or
- operational boundaries.

Lines:

- Shows connections or relationships between launch
- sites and relevant features.
- Enhances understanding of spatial relationships and
- dependencies.

Build a Dashboard with Plotly Dash

Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

Github URL: https://github.com/T-afk-design/courseraCapstone/blob/main/FinalCapstoneDashboard.ipynb

Reasons for Adding Plots and Interactions

Success Pie Chart:

- Provides a quick overview of mission success rates.
- Essential for stakeholders to understand overall performance metrics at a glance.

Success-Payload Scatter Plot:

- Helps identify correlations between payload characteristics and launch outcomes.
- Supports decision-making processes related to payload planning and operational strategies.

Launch Site Dropdown:

- Enhances user experience by focusing analysis on specific launch locations.
- Allows for regional insights and comparisons across different launch sites.

Range Slider for Payload:

- Offers interactive exploration of how payload mass affects mission success.
- Enables detailed analysis and insights into payload-related performance factors.

Predictive Analysis (Classification)

Data Preprocessing:

• Standardized features and split data into training and test sets for model validation.

Model Selection:

• Explored multiple classification algorithms: SVM, Decision Trees, and K-NNearest Neighbors (KNN).

• Chose algorithms suitable for binary classification tasks based on project requirements

Hyperparameter Tuning:

• Used GridSearchCV to systematically search for optimal parameters.

• Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors(KNN).

Model Evaluation:

• Evaluated models using cross-validation techniques to ensure generalization.

• Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

Improvement Iterations:

• Iteratively adjusted models based on insights from validation results.

• Fine-tuned parameters to maximize predictive accuracy.

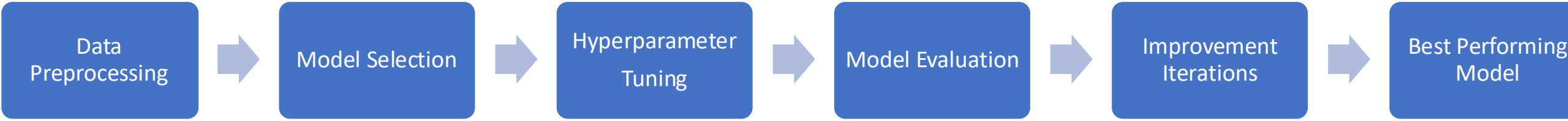
Selection of Best Performing Model:

• Identified the model with the highest accuracy on the test set

• Considered both training and test set performance to avoid overfitting and ensure real-world applicability.

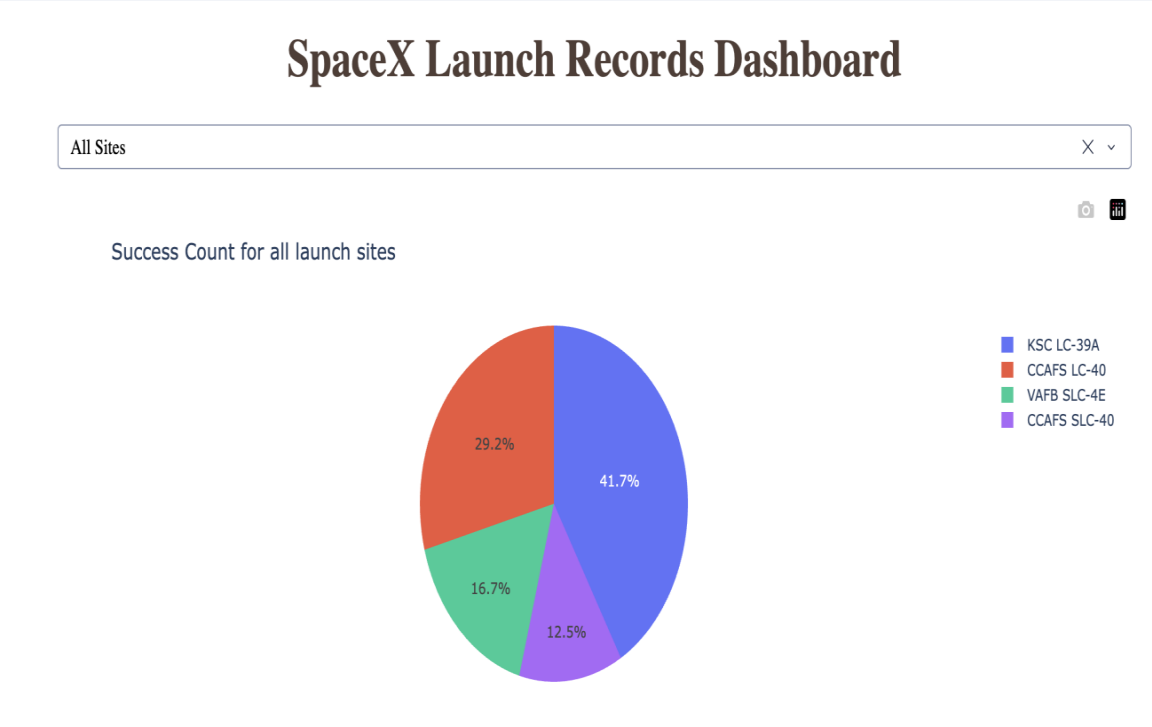
GitHub URL: https://github.com/T-afk-design/courseraCapstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(3).ipynb

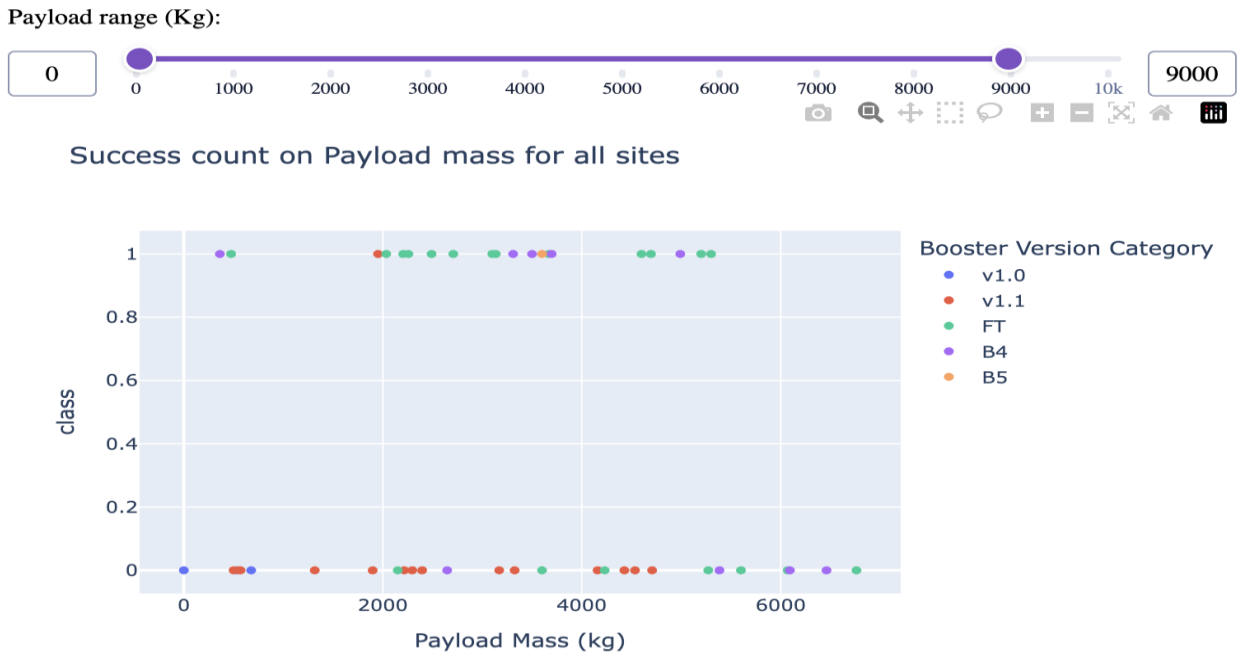
Predictive Analysis (Flowchart)

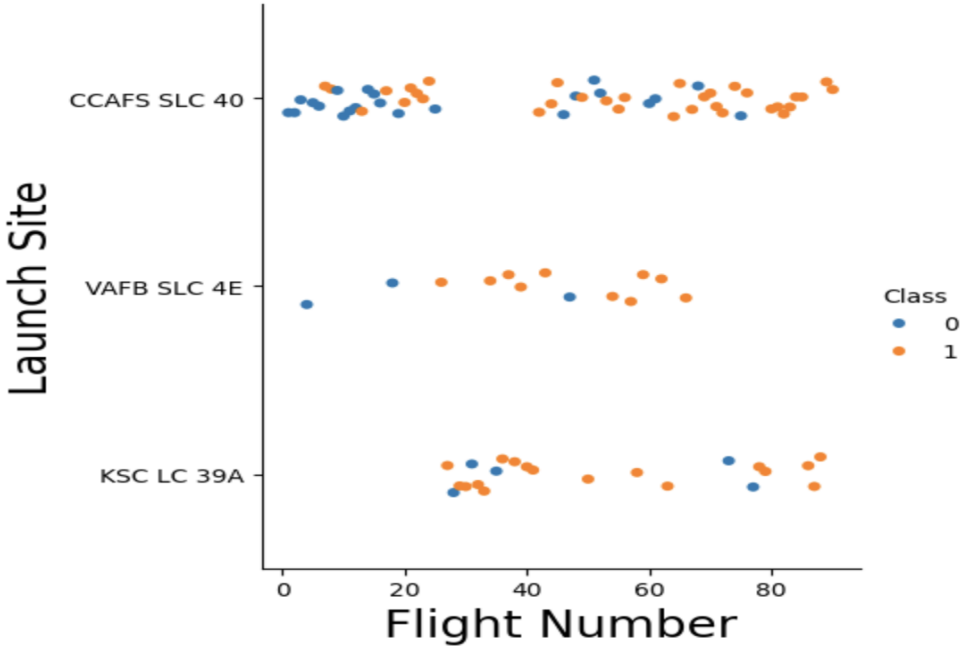


Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results





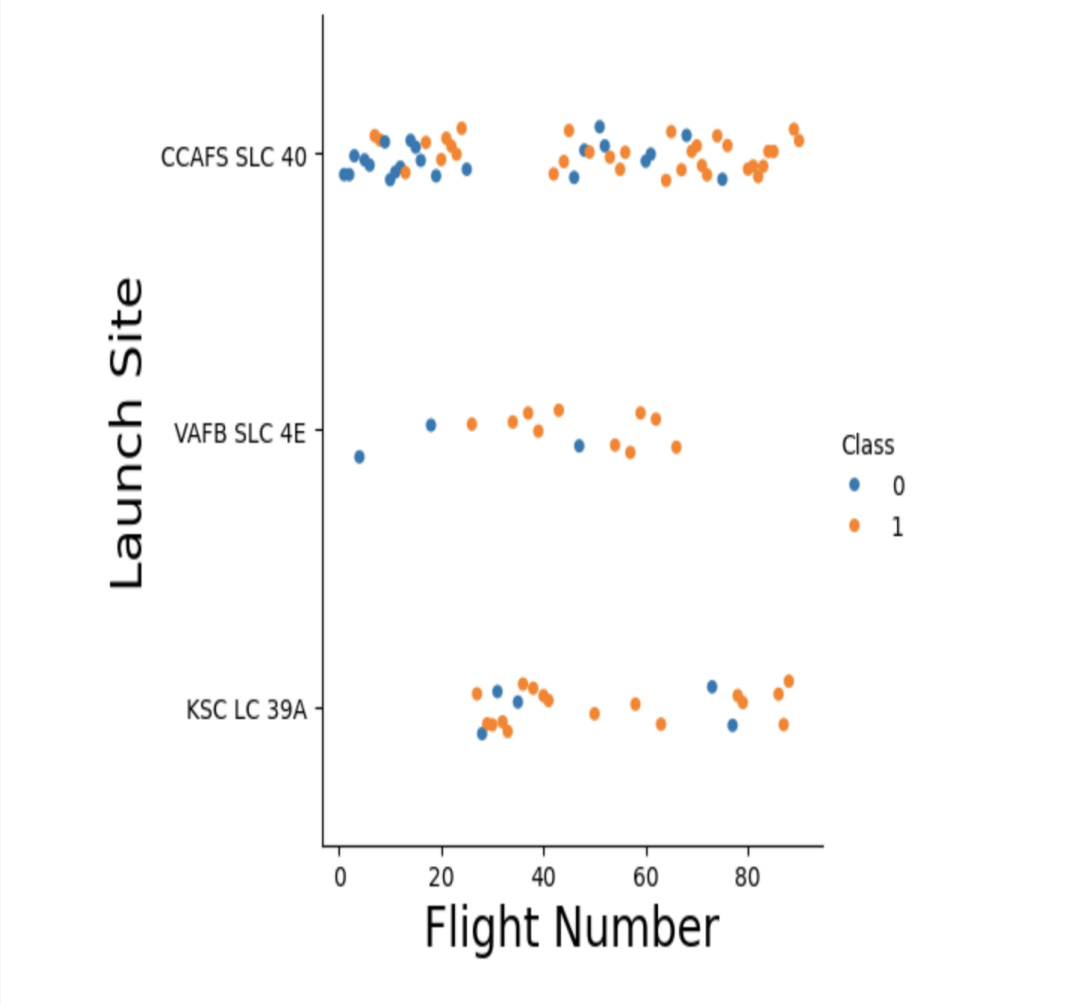


Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

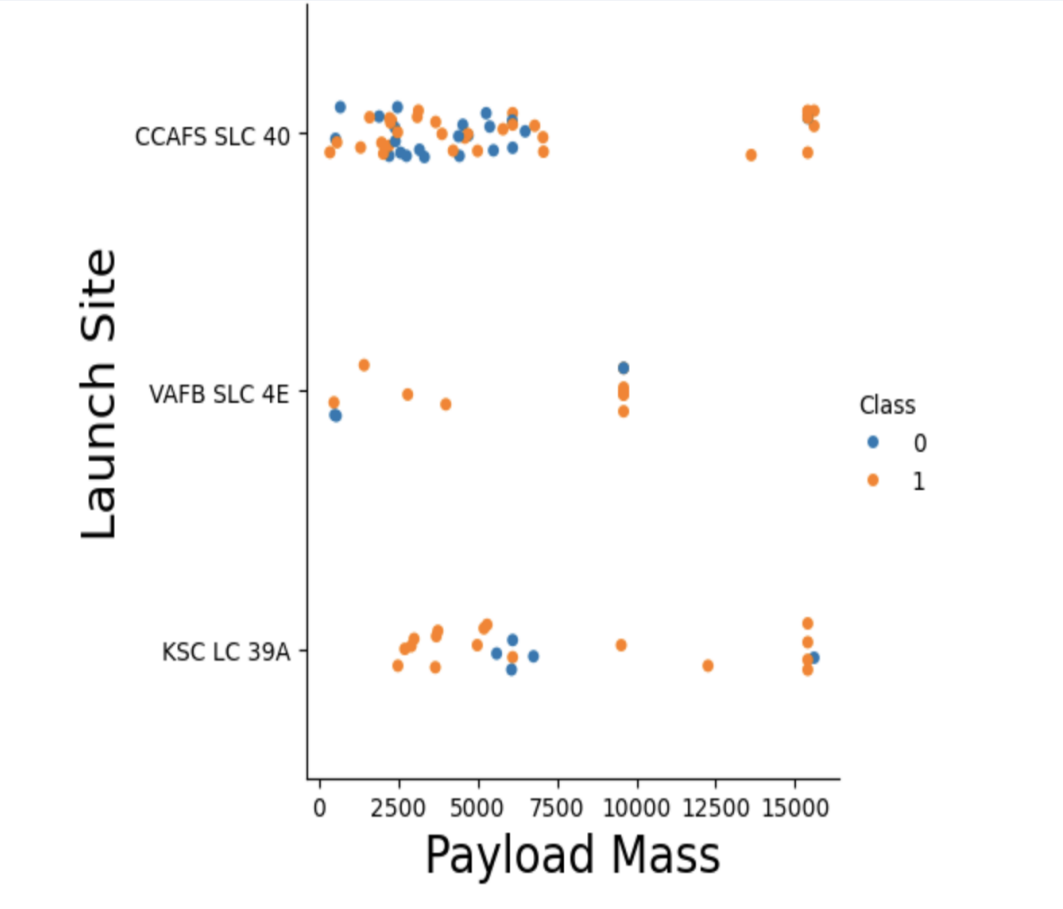
- Scatter plot of Flight Number vs. Launch Site:
  - Both CCAFS SLC 40 and KSC LC 39Ahave a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- Screenshot of the scatter plot with explanations:
  - VAFB SLC 4E had least flights and KSC LC 39A had most flights. Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.



Payload vs. Launch Site

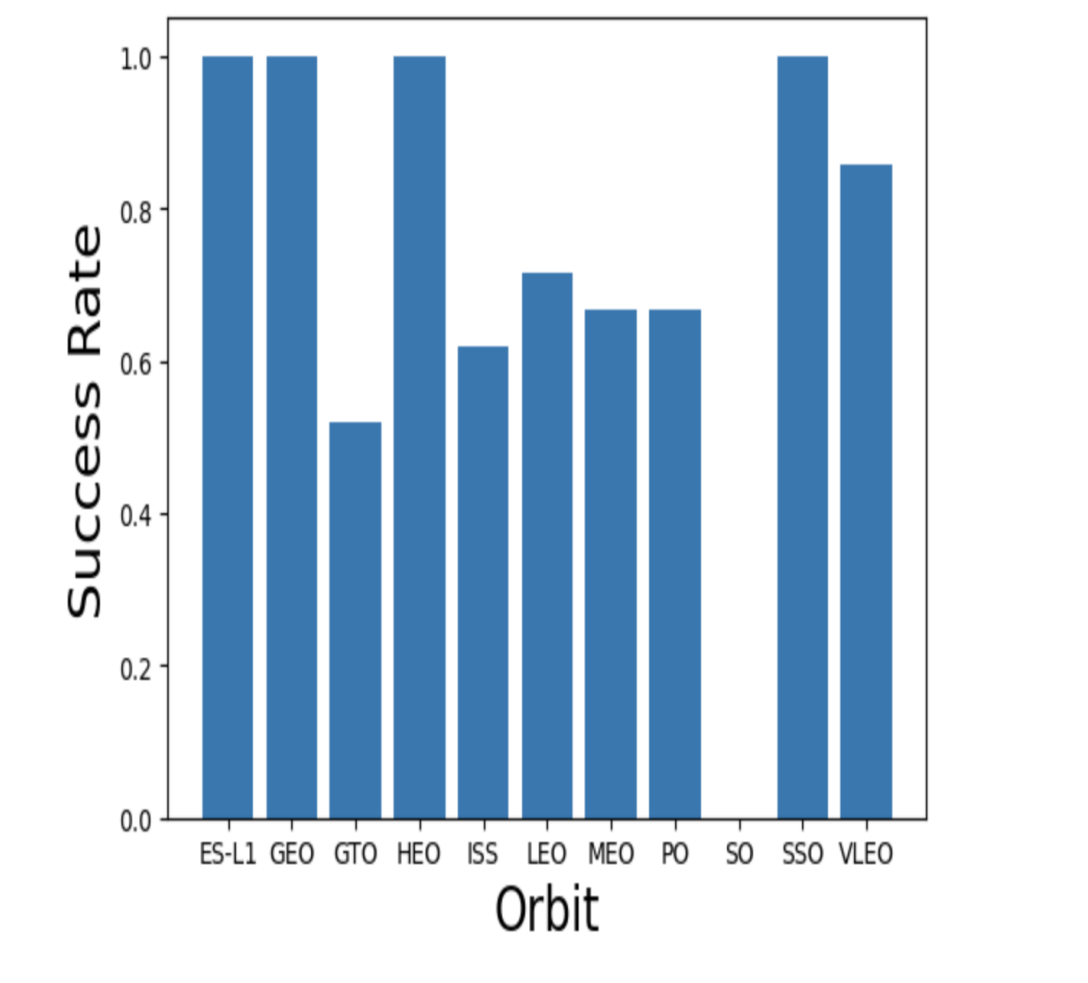
- Scatter plot of Payload vs. Launch Site:

- Screenshot of the scatter plot with explanations:
  - VAFB SLC 4E had least mass. Varied mission profile
  - CCASF SLC 40 had most launches with masses < 8000 units. Mission profile more specific.
  - KSC LC 39A had several successful launches with masses > 15000 units. Varied mission profile. Its heavier payloads suggest suitability for high-capacity missions.



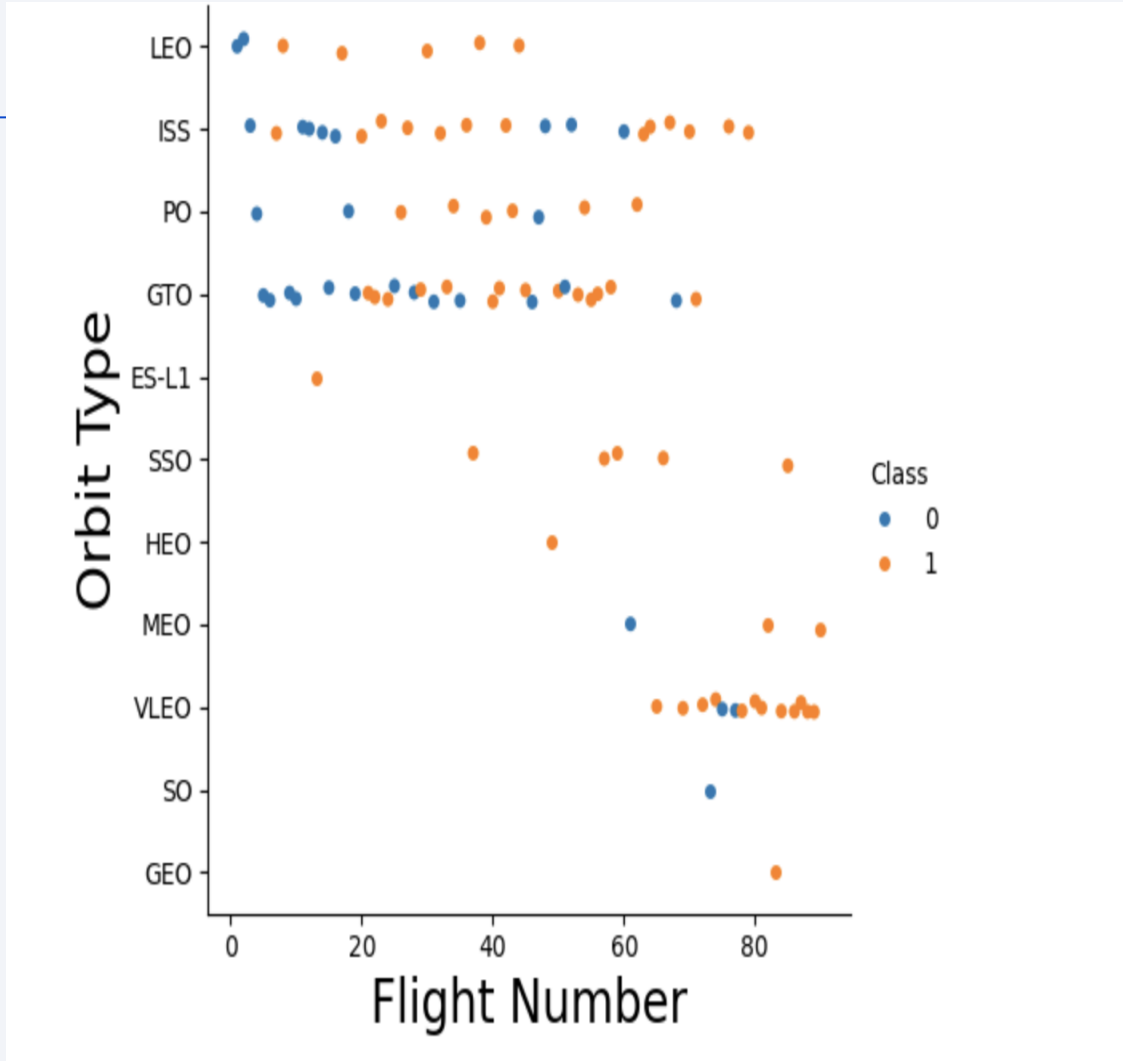
Success Rate vs. Orbit Type

- Bar chart for the success rate of each orbit type:
  - SO orbit type has 0% success rate
  - The success rate for other sites ranges from about 50% to 100%
    - The ES-L1, GEO, HEO, and SSO have 100% rates
      - Indicating these orbits are highly reliable for successful first stage landings.
    - The GTO orbit type shows a significantly lower success rate compared to other orbit types. This indicates that missions to this orbit may involve greater challenges or complexities.



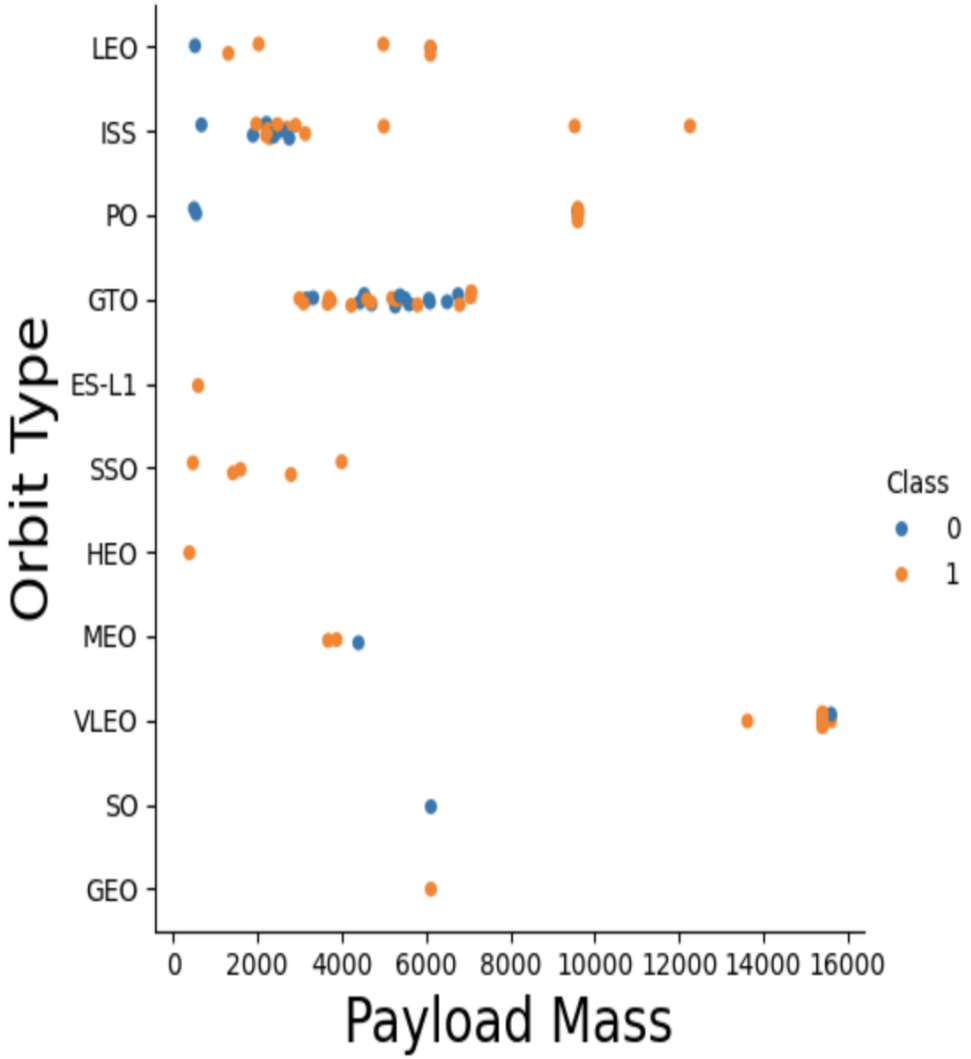
Flight Number vs. Orbit Type

- Scatter plot of Flight number vs. Orbit type
  - Success and failure rates are visualized on the plot.
  - Success and failure rates are correlated with orbit type, and the number of attempted launches
  - Recently, SSO, HEO, MEO, VLEO, SO, and GEO launches have been conducted
  - Increased Success over time
    - Success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
  - Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, showing advancements in mission planning and execution.



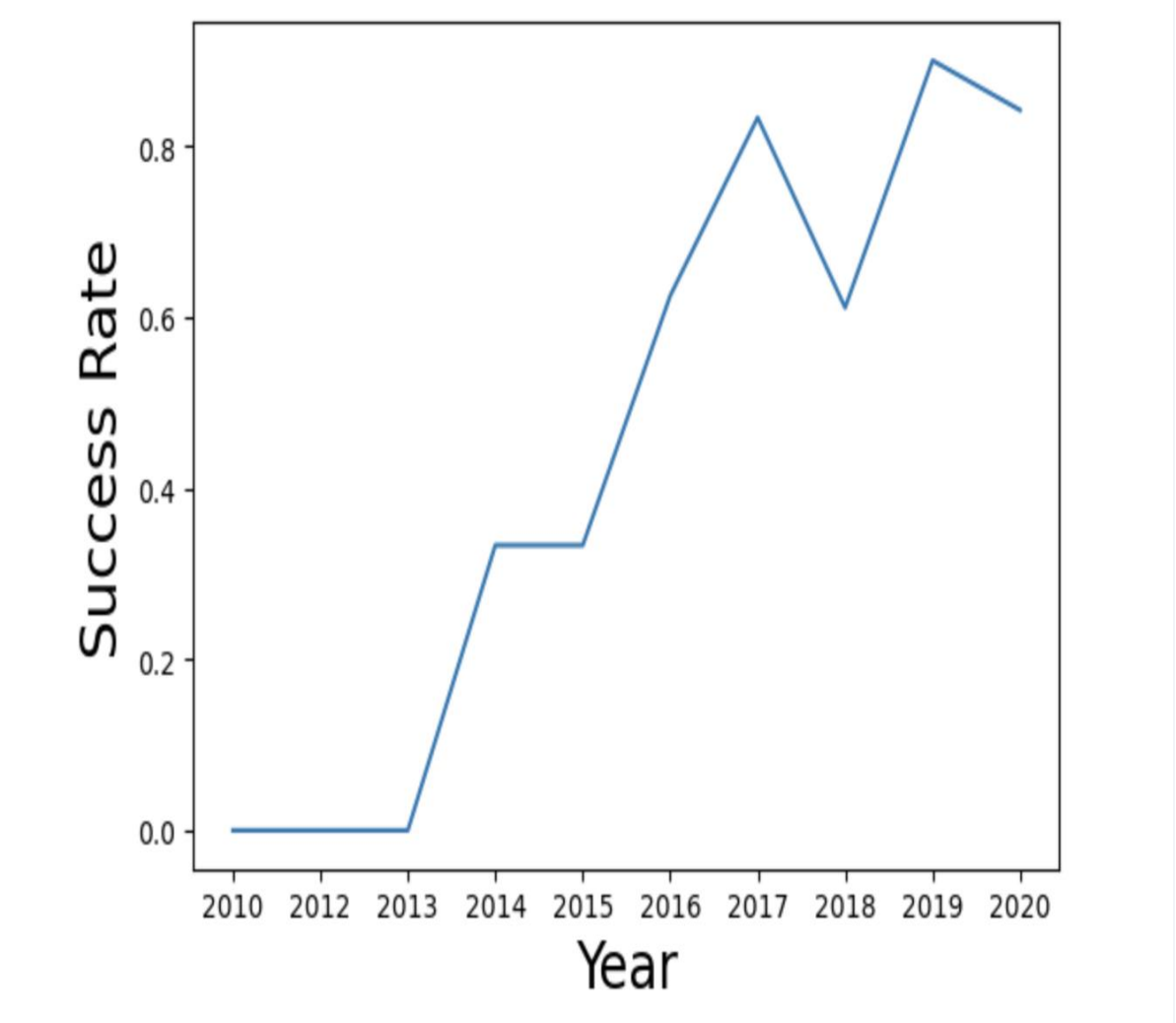
Payload vs. Orbit Type

- Scatter plot of payload vs. orbit type:
  - The heaviest payload masses are handled by VLEO, ISS, and PO
  - The success rate for each orbit type is also shown, with SSO being the most successful repeatedly
  - Successful are more frequent across all orbit types with payloads less than 6000 kg.
  - Higher payload masses (above 10,000 kg) show a mix of successes and failures, indicating increased difficulty with heavier payloads.



Launch Success Yearly Trend

- Line chart of yearly average success rate:
  - Plot shows a sharp increase in success rates over 10 years
  - Success rate was 0% for the first three years
  - Success has reached over 80% in recent years
    - Shows significant improvement from 2013 on onwards
    - There was a success rate dip in 2018



All Launch Site Names

- Find the names of the unique launch sites

%sql SELECT DISTINCT Launch Site FROM SPACEX;

- Present your query result with a short explanation here
  - Launch Site
    - CCAFS LC-40
    - VAFB SLC-4E
    - KSC LC-39A
    - CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`

%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

- query result with a short explanation:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcor
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachu
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachu
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No atten
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No atten
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No atten

Total Payload Mass

- Total payload carried by boosters from NASA

%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';

- Query result:

Display the total payload mass carried by boosters launched by NASA (CRS)

```
2]: %sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';

 * sqlite:///my_data1.db
Done.
```

2]:

Total_Payload_Mass
45596

- Total payload mass is 45596 KG

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1

%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';

- Query result with a short explanation:

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Avg_Payload_Mass
2928.4

- Result: 2928.4 KG

First Successful Ground Landing Date

- Dates of the first successful landing outcome on ground pad

%sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';

- query result with a short explanation:

```
SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

First_Successful_Ground_Pad_Landing
2015-12-22

- Result: 2015-12-2022

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000   AND PAYLOAD_MASS__KG_ < 6000;
```

- query result with a short explanation:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes

%sql select "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";

- query result with a short explanation:

```
%sql select "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WH
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Success	98

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

- Present your query result with a short explanation here

```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- query result with a short explanation:

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
#%sql SELECT substr(Date, 6, 2) AS Month,Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr(Date

%%sql
CASE
WHEN substr("Date", 6, 2) = '01' THEN 'January'
WHEN substr("Date",6,2) = '02' THEN 'February'
WHEN substr("Date", 6,2)='03' THEN 'March'
WHEN substr("Date", 6, 2) = '04' THEN 'April'
WHEN substr("Date", 6, 2) = '05'  THEN 'May'
WHEN substr("Date", 6, 2)='06' THEN 'June'
WHEN substr("Date", 6, 2)= '07' THEN 'July'
WHEN substr("Date", 6, 2)='08' THEN 'August'
WHEN substr("Date", 6, 2)='09' THEN 'September'
WHEN substr("Date", 6, 2)='10' THEN 'October'
WHEN substr("Date", 6, 2)='11' THEN 'November'
WHEN substr("Date", 6, 2)='12' THEN 'December'
ELSE 'Unknown'
END AS "Month_Name",
"Mission_Outcome",
"Booster_Version",
"Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 1, 4) = '2015';
```

E999 SyntaxError: invalid syntax
flake8

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04'
AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

- query result with a short explanation:
  - Ranked in descending order

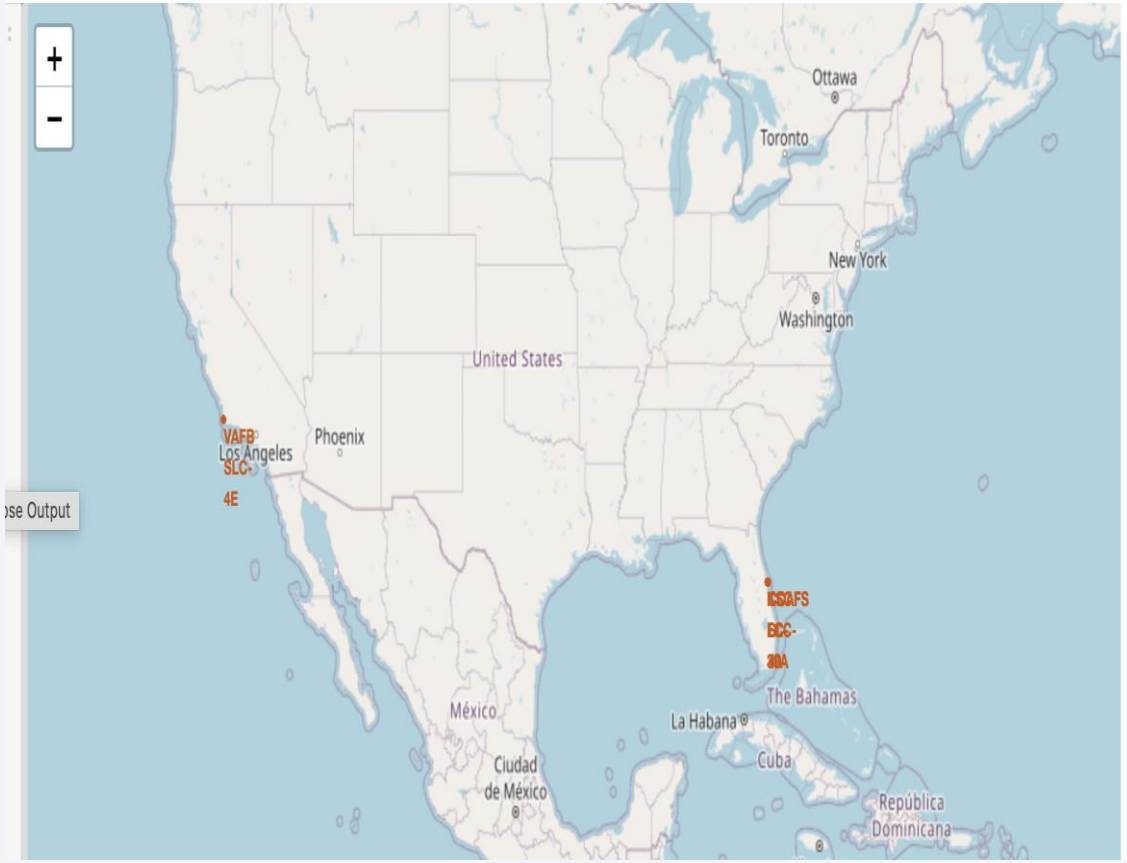
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Folium Map – USA Launch Sites

- Launch sites located in SW and SE corners of continental USA
  - Not all launch sites are in close proximity to the equator
  - The launch site at Vandenberg Air Force Base (VAFB SLC-4E is located at a latitude of 34.63, which is further from the equator compared to the other sites in Florida.
- The launch sites are in red.
- All Launch sites are near the ocean close to the coast. The Falcon 9 can take some advantage of the earth's rotational speed, with somewhat less drag. VAFB SLC-4E is near the coast in California; and KSC LC-39A, CC LC-40, and CCAFS SLC – 40 are near the coast in Florida.

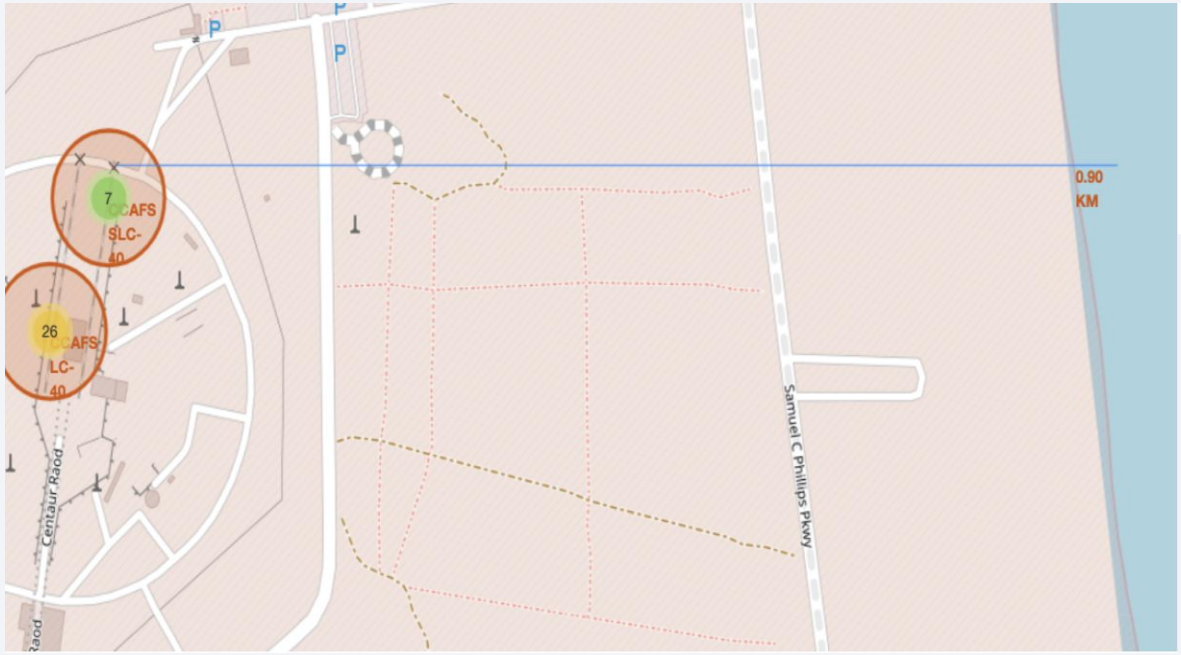


FL coast launch site distance to Orlando

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
  - The proximity of the space station to Orlando allows consolidation of manufacturing, refurbishment, and multiple launch pads in one area. This location allows Florida to be a central hub for space exploration

Visual scaled distance to coastline

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot
- There are two launchpads, one is slightly closer to the coast than the other. Both are between 0.8 and 1.6 km from the coast




```
distance_coastline = calculate_distance(28.575,-80.658,28.575 ,-80.645)
distance_coastline
```

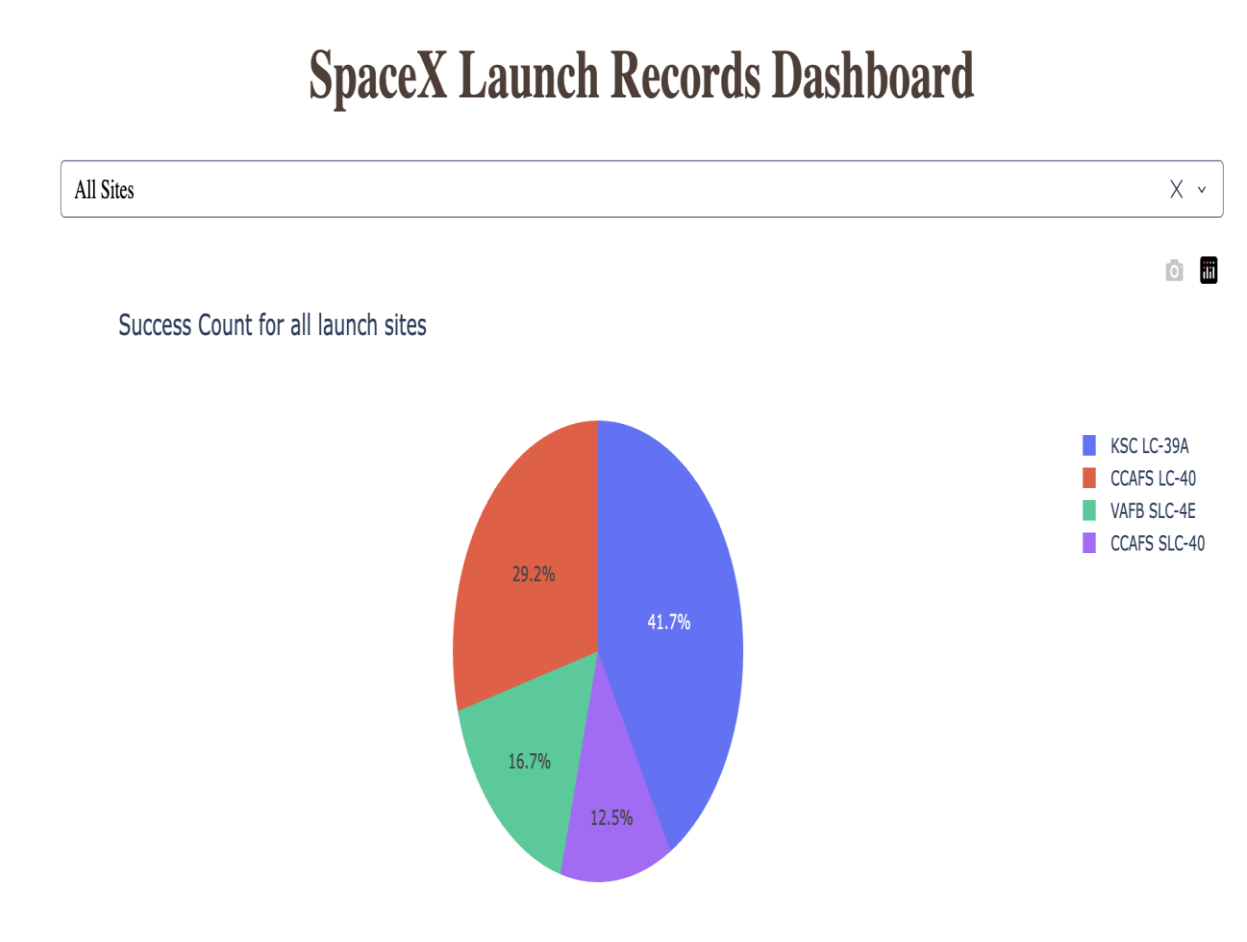
```
]: 1.2698545993095536
```

Section 4

Build a Dashboard with Plotly Dash

Dashboard Screenshot: Launch Success Count for All Sites

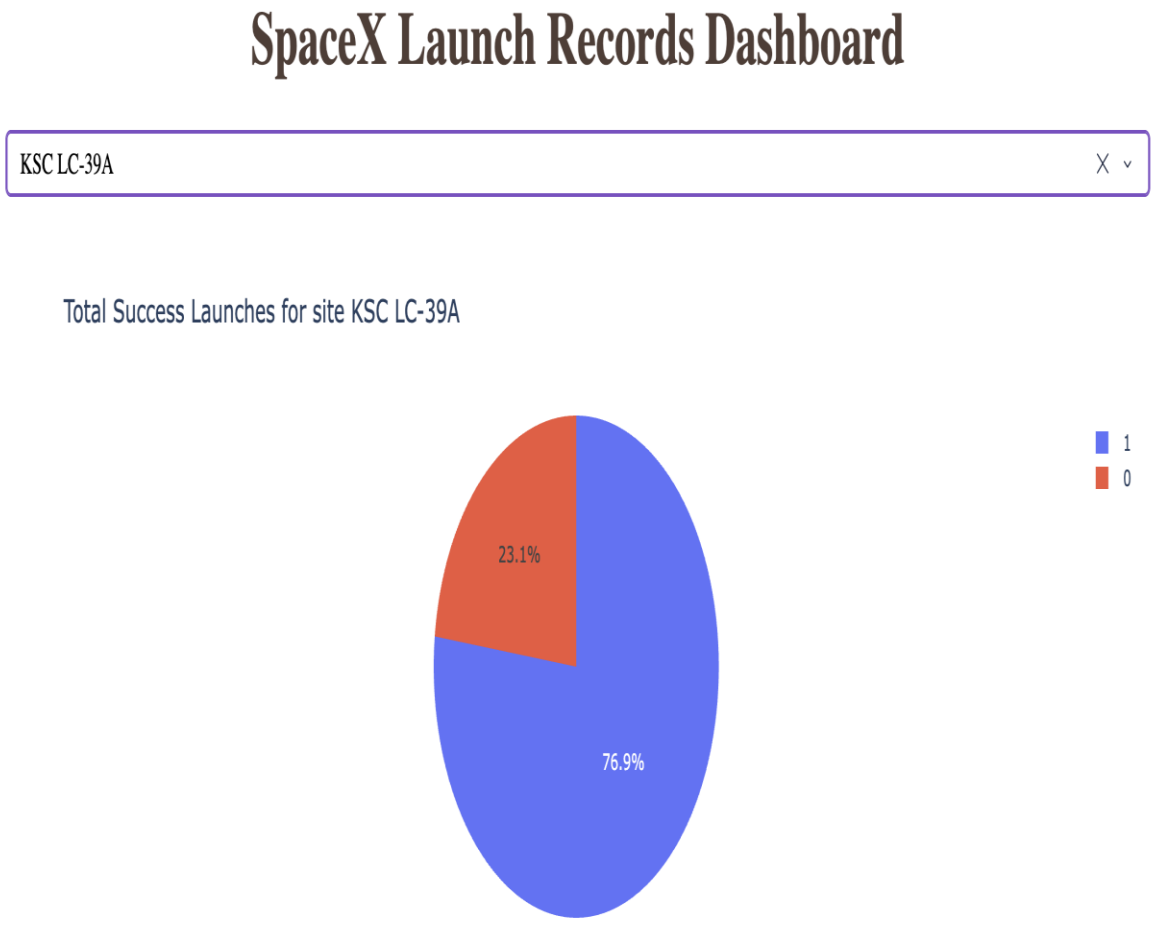
- The site with the most successful launches was KSC LC-39A (41.7%). This indicates this site is a highly reliable site for SpaceX launches.
- The site with the fewest successful launches was CCAFS SLC-40 (12.5%)
- CCAFS LC-40 had 29.2% success rate
- VAFB SLC–4E had 16.7% success rate



Dashboard Screenshot: Site with Highest Success Launch Ratio

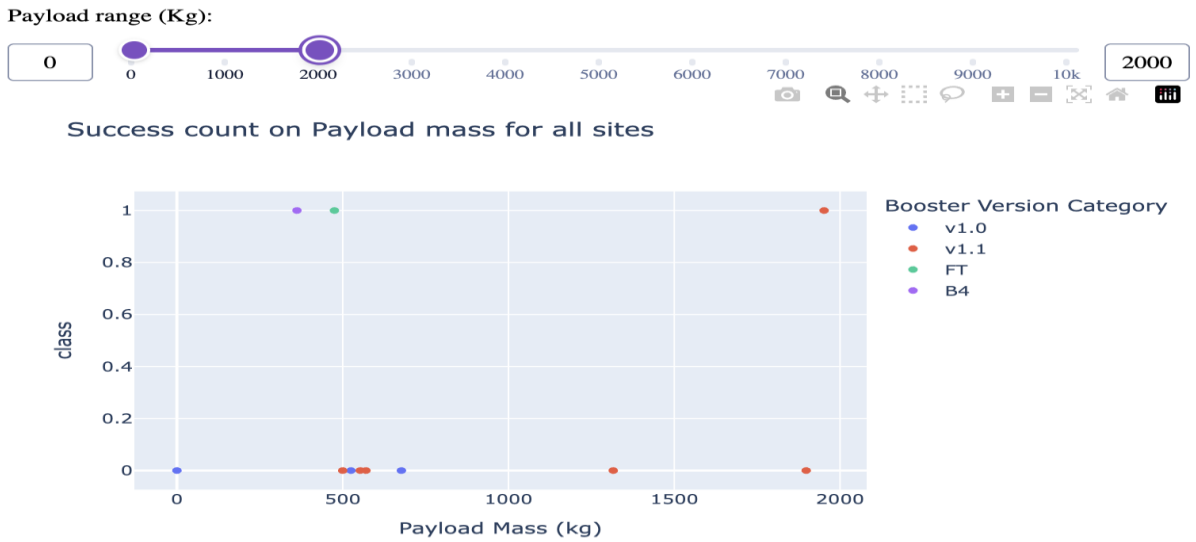
- Piechart for the launch site with highest launch success ratio

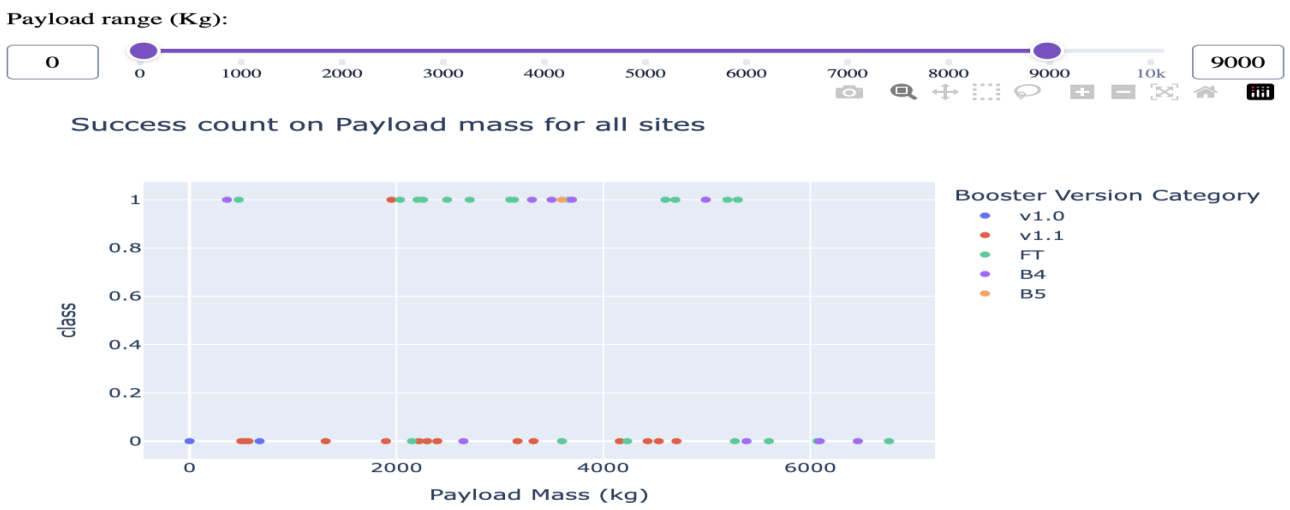
  - KSC LC-39A has the highest success rate of 76.9%. Its failure rate is 23.1%. This indicates this site is a highly reliable site for SpaceX launches.



Dashboard Screenshot: Payload vs. Launch Outcome scatter plot

- Payload vs. Launch Outcome scatter plot for all sites, with different payloads selected





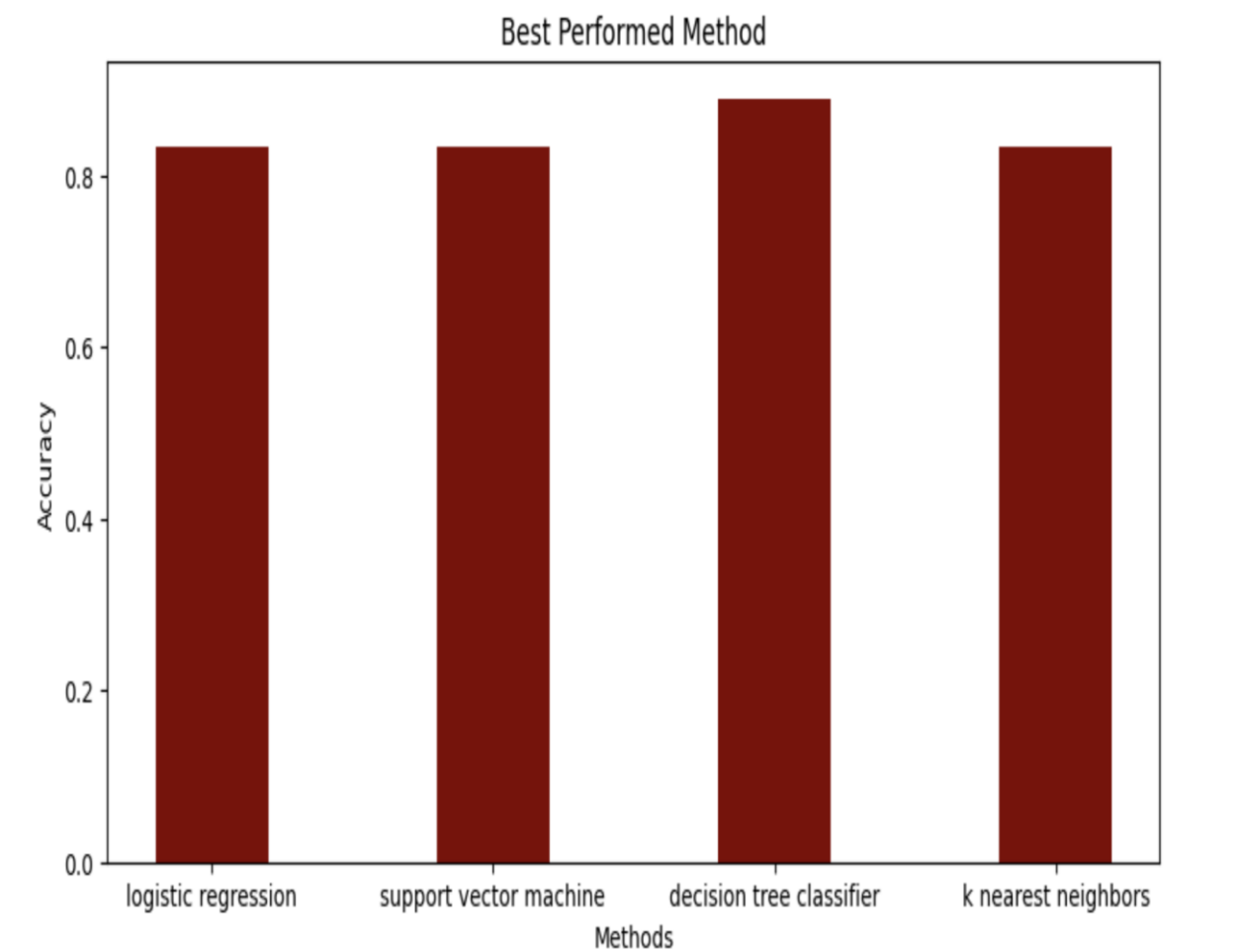
- There are more failure rates for all Booster versions if the load is above 5K KG. The trend is vague, but there could be a correlation.
- The FT Booster version has the highest success rate for all loads, and is most frequently used.
- The v1.1 Booster version has the highest failure rate for all loads Booster version v1.0 has fewer launches and requires further analysis to understand its performance.
- CCAFS LC-40 has the highest success rate, so it is the most reliable launch site.
- The other sites have variable success launch rates.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualized the built model accuracy for all built classification models in a bar chart:
  - Based on the results and the number of times I ran the testing, Decision Tree model has the highest classification accuracy on the test data set. Its accuracy achieved 0.88 or higher when I ran the program. This implies that Decision Tree Model is better suited for this dataset compared to logistic regression, support vector machine, and k nearest neighbors, all of which achieved an accuracy of just 0.83.



Confusion Matrix

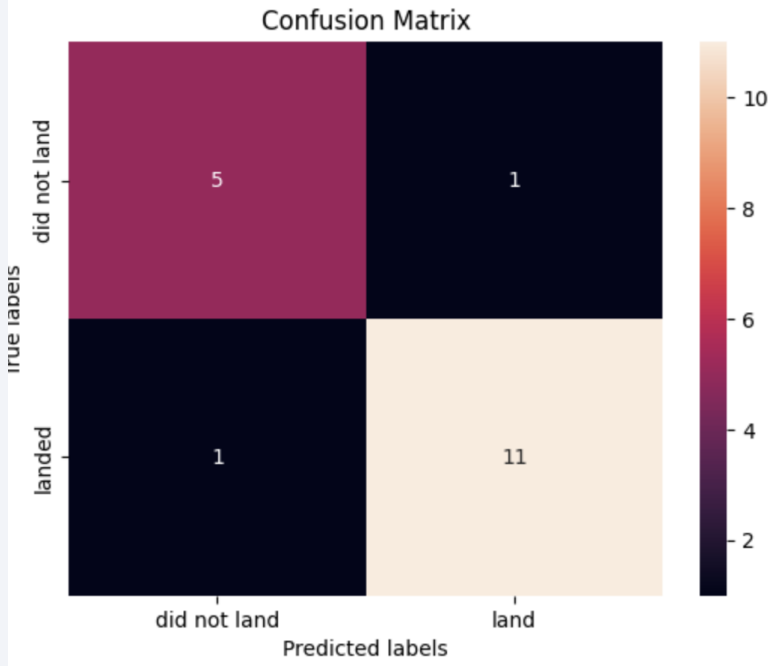
Explanation and Insights

- **High Accuracy**: The model achieved a high accuracy score of 88.88%, with a significant number of true positives and true negatives, demonstrating its effectiveness in predicting Falcon 9 first stage landings.
- **No False Negatives**: The absence of false negatives indicates that the model reliably predicts successful landings. This is crucial for ensuring readiness and safety in aerospace operations, as every actual successful landing was accurately identified.
- **Manageable False Positives**: While there are 2 false positives, this is less critical than false negatives in aerospace operations. Over-preparation (due to false positives) is more manageable than under-preparation, making the model's performance highly acceptable for practical applications.
- **Balanced Performance**: The model shows a balanced performance with a slight bias towards predicting successful landings. This aligns well with practical needs in the aerospace industry, where ensuring successful landings is of paramount importance for cost estimation and planning.

```
0.8888888888888888
```

We can plot the confusion matrix

```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



Conclusions

- The CCAFS LC-40 launch site has the highest success rate among all sites, accounting for 43.7% of successful launches. This indicates that this site might have optimal conditions or processes that contribute to a higher success rate.
- The scatter plot analysis showed that the “FT” booster version has a high success rate across various payload masses. This demonstrates reliability and robustness compared to other booster versions. This suggests that future missions might benefit from utilizing this booster version for improved success rates.
- Success rates may be correlated with lower payload mass, specific launch sites, booster version, and mission type. How payload mass has the least correlation with success rates suggesting other factors play a larger role.
- Predictive analysis is highly confident of future success rates. Predictive analysis with Folium and Plotly Dash have allowed for a deep understanding of the data. These tools allow for insight into geographical and operational patterns
- In conclusion, the insights gathered can help improve launch strategies and contribute to reusable rocket technology. The data analyzed has answered questions on factors that influence SpaceX launch success rates. The data gathered can be used to improve future launch success strategies.

Appendix

- All relevant code and outputs can be found here:

  https://github.com/T-afk-design/courseraCapstone

Thank you!